Analysis I - Problem Set 1

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Exercise 1: Let s be an upper bound of a non-empty subset $E \subset \mathbb{R}$. Then show that s is the least upper bound of E if and only if for every $\epsilon > 0$, there is $u_\epsilon \in E$ such that $s - \epsilon < u_\epsilon$.

Proof. We show both directions seperately.
($\rightarrow$) Suppose $s = \sup E$ and $\epsilon > 0$ be given. Since $s - \epsilon < s$ and s is the least upper bound, $s - \epsilon$ cannot be an upper bound of E. Therefore, there exists some $u_\epsilon \in E$ such that $u_\epsilon > s - \epsilon$.
($\leftarrow$) Suppose for every $\epsilon > 0$, there exists $u_\epsilon \in E$ such that $s - \epsilon < u_\epsilon$. We need to show that s is the least upper bound of E. Let v be an upper bound of E. Then for every $\epsilon > 0$, we have $u_\epsilon \leq v$. This implies that $s - \epsilon < v$ for every $\epsilon > 0$, thus we get $s \leq v$ [see proof for this after the proof of this exercise is completed]. Since s is an upper bound of E and $s \leq v$ for any upper bound v, it follows that s is the least upper bound of E. $\square$

Theorem I. *If for every $\epsilon > 0$, we have $x - \epsilon < y$ for some $x, y \in \mathbb{R}$, then $x \leq y$.*

Proof. Assume the contrary that $x > y$. Then, we have:

$$x - y > 0$$

Let $\epsilon = \frac{x-y}{2}$. Note that $\epsilon > 0$. Then, we have:

$$x - \epsilon = x - \frac{x-y}{2} = \frac{x+y}{2} > y$$

This is a contradiction since we assumed that for every $\epsilon > 0$, we have $x - \epsilon < y$. Hence, we conclude that our assumption is false. Therefore, we have $x \leq y$. $\square$

Corollary I.1:

If for every $\epsilon > 0$, we have $x < \epsilon$ for some $x \geq 0$, then $x = 0$.

Proof. From the above theorem, we have $x \leq 0$ since $x < \epsilon \implies x - \epsilon < 0$. Since $x \geq 0$, it follows that $x = 0$. $\square$

Exercise 2: Formulate -

1. A definition of a lower bound of a non-empty set $E \subset \mathbb{R}$

2. A definition for the greatest lower bound of a non-empty bounded below subset E of $\mathbb{R}$.

Proof.
1. A number $m \in \mathbb{R}$ is called a lower bound of a non-empty set $E \subset \mathbb{R}$ if for every $x \in E$, we have $m \leq x$.

2. A number $g \in \mathbb{R}$ is called the greatest lower bound (or infimum) of a non-empty bounded below subset E of $\mathbb{R}$ if:

   - g is a lower bound of E, i.e., for every $x \in E$, we have $g \leq x$.
   - For any other lower bound m of E, we have $m \leq g$. In other words, g is the largest among all lower bounds of E.

We can alternatively define the greatest lower bound using the following property: A number $g \in \mathbb{R}$ is the greatest lower bound of E if for every $\epsilon > 0$, there exists an element $x_\epsilon \in E$ such that $g \leq x_\epsilon < g + \epsilon$. $\square$

Exercise 3: Show that every non-empty bounded below subset of the real numbers admit a greatest lower bound. The greatest lower bound of a non-empty bounded below set E is denoted by $\inf E$.

Proof. Let $A \subset \mathbb{R}$ be a non-empty bounded below subset. Then it follows that -

$$\exists m \in \mathbb{R} : m \leq x, \forall x \in A$$

Define the set $B = \{-x : x \in A\}$. Since A is bounded below, it follows that B is bounded above by $-m$ -

$$m \leq x \implies -x \leq -m \implies -m \text{ is an upper bound of } B$$

Since A is non-empty, B is also non-empty. By the completeness property of real numbers, every non-empty bounded above subset of $\mathbb{R}$ has a least upper bound. Therefore, B has a least upper bound, say $s = \sup B$. We claim that $-s$ is the greatest lower bound of A. To see this, we need to show two things:

- $-s$ is a lower bound of A.
- For any other lower bound m of A, we have $m \leq -s$.

Proof that $-s$ is a lower bound of A: Let $x \in A$. Then $-x \in B$. Since s is an upper bound of B, we have $-x \leq s$. This implies that $x \geq -s$. Since this holds for all $x \in A$, it follows that $-s$ is a lower bound of A.
Proof that $-s$ is the greatest lower bound of A: Let m be any lower bound of A. Then for every $x \in A$, we have $m \leq x$. This implies that for every $y \in B$ (where $y = -x$ for some $x \in A$), we have $-y \leq -m$, or equivalently, $y \geq -m$. Therefore, $-m$ is an upper bound of B. Since s is the least upper bound of B, we have $s \leq -m$. This implies that $-s \geq m$. Since this holds for any lower bound m of A, it follows that $-s$ is the greatest lower bound of A. Thus, we conclude that every non-empty bounded below subset of the real numbers admits a greatest lower bound. □

Exercise 4: Formulate a definition for the greatest lower bound of a set in the context of exercise 1.

Proof. A number $g \in \mathbb{R}$ is the greatest lower bound of E if for every $\epsilon > 0$, there exists an element $x_\epsilon \in E$ such that $g \leq x_\epsilon < g + \epsilon$. □

Exercise 5: Let $A = \{\frac{1}{n} : n \in \mathbb{N}\}$. Find $\sup A$.

Proof. We claim that $\sup A = 1$. To see this, first note that 1 is an upper bound of A since for every $n \in \mathbb{N}$, we have $\frac{1}{n} \leq 1$. Now, let $\epsilon > 0$ be given. We need to show that there exists an element $u_\epsilon \in A$ such that $1 - \epsilon < u_\epsilon$. Since 1 is a upper bound, $\sup A > 1$ is not possible. We show now that $\sup A < 1$ is also not possible. Note that for $n = 1$ (which is in $\mathbb{N}$), and hence $1 \in A$. Therefore $\sup A < 1$ is a contradiction by the definition of the supremum. Hence, we conclude that $\sup A = 1$. □

Exercise 6: Let A, B be non-empty subsets of $\mathbb{R}$ such that $a \leq b$ for all $a \in A$ and $b \in B$. Let A be bounded above and B be bounded below. Show that $\sup A \leq \inf B$.

Proof. Let $s = \sup A$ and $i = \inf B$. We need to show that $s \leq i$. Since s is an upper bound of A, we have $a \leq s$ for all $a \in A$. Since i is a lower bound of B, we have $i \leq b$ for all $b \in B$. Suppose for the sake of contradiction that $s > i$. Then, by the definition of infimum, we have -

$$\forall \epsilon > 0, \exists b_\epsilon \in B : i \leq b_\epsilon < i + \epsilon$$

Let us take $\epsilon = s - i$. Then, we have:

$$\exists b_\epsilon \in B : i \leq b_\epsilon < i + (s - i) \implies i \leq b_\epsilon < s$$

Since $a \leq b$ for all $a \in A$ and $b \in B$, we have $a \leq b_\epsilon$ for all $a \in A$. This implies that b_ϵ is an upper bound of A. Since s is the least upper bound of A, we have $s \leq b_\epsilon$. This is a contradiction since we have $b_\epsilon < s$. Hence, we conclude that our assumption that $s > i$ is false. Therefore, we have $s \leq i$, or equivalently, $\sup A \leq \inf B$. □

Exercise 7: Let A, B be non-empty bounded subsets of $\mathbb{R}$. Let $A + B = \{a + b : a \in A, b \in B\}$. Show that $\sup(A + B) = \sup A + \sup B$.

Proof. Let $s_A = \sup A$ and $s_B = \sup B$. We need to show that $\sup(A+B) = s_A + s_B$. First, we show that $s_A + s_B$ is an upper bound of $A+B$. Let $x \in A+B$. Then, by definition, there exist $a \in A$ and $b \in B$ such that $x = a+b$. Since s_A is an upper bound of A, we have $a \leq s_A$. Similarly, since s_B is an upper bound of B, we have $b \leq s_B$. Adding these two inequalities, we get:

$$x = a + b \leq s_A + s_B$$

This shows that $s_A + s_B$ is an upper bound of $A+B$.
Now, we need to show that $s_A + s_B$ is the least upper bound of $A+B$. We write the definition of supremums for A and B:

1. For every $\epsilon > 0$, there exists $a_\epsilon \in A$ such that
$$s_A - \frac{\epsilon}{2} < a_\epsilon \leq s_A \tag{1}$$
2. For every $\epsilon > 0$, there exists $b_\epsilon \in B$ such that
$$s_B - \frac{\epsilon}{2} < b_\epsilon \leq s_B \tag{2}$$

Adding (1) and (2), we get:
$$s_A + s_B - \epsilon < a_\epsilon + b_\epsilon \leq s_A + s_B$$
Note that $a_\epsilon + b_\epsilon \in A+B$. Therefore, for every $\epsilon > 0$, there exists an element $x_\epsilon = a_\epsilon + b_\epsilon \in A+B$ such that:
$$s_A + s_B - \epsilon < x_\epsilon \leq s_A + s_B$$
Hence, $s_A + s_B = \sup(A+B) = \sup A + \sup B$ □

Exercise 8: Prove that -

1. $\sup\{1 - \frac{1}{n} : n \in \mathbb{N}\} = 1$
2. $\inf\{\frac{1}{n^2} : n \in \mathbb{N}\} = 0$

Proof. We prove both parts seperately,

1. Let $A = \{1 - \frac{1}{n} : n \in \mathbb{N}\}$. We claim that $\sup A = 1$. To see this, first note that 1 is an upper bound of A since for every $n \in \mathbb{N}$, we have $1 - \frac{1}{n} < 1$. Now, let $\epsilon > 0$ be given. We need to show that there exists an element $u_\epsilon \in A$ such that $1 - \epsilon < u_\epsilon$. Since 1 is an upper bound, $\sup A > 1$ is not possible. We show now that $\sup A < 1$ is also not possible. Suppose for the sake of contradiction that $\sup A = u < 1$. Then $\forall n \in \mathbb{N}$, we have $1 - \frac{1}{n} \leq u < 1$. This implies that $\frac{1}{n} \geq 1 - u > 0$ for all $n \in \mathbb{N}$. This implies that $1 \geq n(1-u)$ for all $n \in \mathbb{N}$. Since $1 - u > 0$, this is a contradiction since we can always choose n large enough such that $n(1-u) > 1$ by the archimedean property of real numbers. Hence, we conclude that our assumption that $\sup A < 1$ is false. Therefore, we have $\sup A = 1$.
2. Let $B = \{\frac{1}{n^2} : n \in \mathbb{N}\}$. We claim that $\inf B = 0$. To see this, first note that 0 is a lower bound of B since for every $n \in \mathbb{N}$, we have $\frac{1}{n^2} > 0$. Now, let $\epsilon > 0$ be given. We need to show that there exists an element $v_\epsilon \in B$ such that $v_\epsilon < \epsilon$. Since 0 is a lower bound, $\inf B < 0$ is not possible. We show now that $\inf B > 0$ is also not possible. Suppose for the sake of contradiction that $\inf B = l > 0$. Then $\forall n \in \mathbb{N}$, we have $\frac{1}{n^2} \geq l > 0$. This implies that $n^2 \leq \frac{1}{l}$ for all $n \in \mathbb{N}$. Since $l > 0$, this is a contradiction since we can always choose n large enough such that $n^2 > \frac{1}{l}$ by the archimedean property of real numbers. Hence, we conclude that our assumption that $\inf B > 0$ is false. Therefore, we have $\inf B = 0$.

□

Exercise 9: Show that $\inf\{n + \frac{1}{m} : n, m \in \mathbb{N}\} = 1$.

Proof. Let $A = \{n : n \in \mathbb{N}\}$ and $B = \{\frac{1}{m} : m \in \mathbb{N}\}$. Then, we can write the given set as $A+B = \{n + \frac{1}{m} : n, m \in \mathbb{N}\}$. We know that $\inf A = 1$ and $\inf B = 0$ from previous exercises. From a previous result, we have $\inf(A+B) = \inf A + \inf B = 1 + 0 = 1$. Hence, we conclude that $\inf\{n + \frac{1}{m} : n, m \in \mathbb{N}\} = 1$. □

<u>Exercise 10:</u> Find the infinimum of the set $\{n + \frac{1}{n} : n \in \mathbb{N}\}$

Proof. We claim that the infimum is 2. To see this, first note that for every $n \in \mathbb{N}$, we have:

$$n + \frac{1}{n} \geq 2$$

This shows that 2 is a lower bound of the set. Now, let $\epsilon > 0$ be given. We need to show that there exists an element x_ϵ in the set such that:

$$2 \leq x_\epsilon < 2 + \epsilon$$

Suppose there exists no such element. Then, we have:

$$n + \frac{1}{n} \geq 2 + \epsilon, \forall n \in \mathbb{N}$$

This implies that:

$$n^2 - (2 + \epsilon)n + 1 \geq 0, \forall n \in \mathbb{N}$$

The roots of the quadratic equation $n^2 - (2 + \epsilon)n + 1 = 0$ are given by:

$$n = \frac{(2 + \epsilon) \pm \sqrt{(2 + \epsilon)^2 - 4}}{2}$$

For the quadratic to be non-negative for all $n \in \mathbb{N}$, the discriminant must be less than or equal to zero:

$$(2 + \epsilon)^2 - 4 \leq 0$$

$$\epsilon^2 + 4\epsilon \leq 0$$

This is a contradiction since $\epsilon > 0$. Hence, we conclude that our assumption is false. Therefore, we have shown that for every $\epsilon > 0$, there exists an element x_ϵ in the set such that:

$$2 \leq x_\epsilon < 2 + \epsilon$$

Here we assume the existence of the square root function. We prove that such a function exists in the next exercise. Hence, we conclude that the infimum of the set $\{n + \frac{1}{n} : n \in \mathbb{N}\}$ is 2. □

<u>Exercise 11:</u> Let $a > 0$ be a given real number. Then show that there exists a unique positive number $x \in \mathbb{R}$ such that $x^2 = a$.

Proof. We prove first that such a number exists. Let $S = \{x \in \mathbb{R} : x^2 < a\}$. Note that $0 \in S$ since $0^2 = 0 < a$. Hence, S is non-empty. We now show that S is bounded above. Note that for any $x > a+1$, we have:

$$x^2 > (a + 1)^2 > a$$

Hence, $a + 1$ is an upper bound of S. By the completeness property of real numbers, every non-empty bounded above subset of $\mathbb{R}$ has a least upper bound. Therefore, S has a least upper bound, say $x = \sup S$. We claim that $x^2 = a$. To see this, we consider the following cases:

1. If $x^2 < a$, then we have:

   $$a - x^2 > 0$$

   Let $\epsilon = \min\{1, \frac{a-x^2}{2x+1}\}$. Note that $\epsilon > 0$ since $x \geq 0$. Then, we have:

   $$(x + \epsilon)^2 = x^2 + 2x\epsilon + \epsilon^2 < x^2 + (2x + 1)\epsilon = x^2 + (a - x^2) = a \qquad [\epsilon^2 < \epsilon \text{ as } \epsilon < 1]$$

   This implies that $x + \epsilon \in S$. But this contradicts the fact that x is an upper bound of S. Hence, this case is not possible.

2. If $x^2 > a$, then we have:

   $$x^2 - a > 0$$

   Let $\epsilon = \min\{1, \frac{x^2-a}{2x}\}$. Note that $\epsilon > 0$ since $x > 0$. Then, we have:

   $$(x - \epsilon)^2 = x^2 - 2x\epsilon + \epsilon^2 > x^2 - 2x\epsilon = x^2 - (x^2 - a) = a$$

   This implies that $x - \epsilon$ is an upper bound of S which is less than x. But this contradicts the fact that x is the least upper bound of S. Hence, this case is not possible.

Therefore, the only possibility left is that $x^2 = a$. Hence, we have shown that there exists a positive number $x \in \mathbb{R}$ such that $x^2 = a$.
We now show that such a number is unique. Suppose there exist two positive numbers $x_1, x_2 \in \mathbb{R}$ such that $x_1^2 = a$ and $x_2^2 = a$. Then, we have:

$$x_1^2 = x_2^2 \implies (x_1 - x_2)(x_1 + x_2) = 0$$

Since $x_1, x_2 > 0$, we have $x_1 + x_2 > 0$. Hence, we must have $x_1 - x_2 = 0$. This implies that $x_1 = x_2$. Thus, we have shown that the positive number $x \in \mathbb{R}$ such that $x^2 = a$ is unique. $\square$